ATUL RAJ

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EDUCATION

NIST University, Berhampur, Odisha Nov 2022 – Apr 2026 (expected)
B.Tech in Computer Science and Engineering CGPA: 8.5

JVM Shyamali, Ranchi 2022
Class XII 72%

D.A.V. Kapildev, Ranchi 2020
Class X 94.5%

EXPERIENCE

Research Intern, Defence Research and Development Organisation (DRDO) June 2025 – August 2025

- Researched Quantum Computing & QML (QFT, QPE, Grover's, Shor's) for cryptography and optimization.
- Simulated quantum circuits using IBM Qiskit to evaluate performance across real-world use cases.
- Explored applications of quantum algorithms in secure communication and complex problem-solving.

PROJECTS

JournEaze – AI-Powered Collaborative Travel Platform Live | GitHub

- Built real-time collaborative travel platform with shared workspaces, drag-drop planning, and Mapbox routing.
- Developed Groq AI features including Smart Packing Lists and dynamic Budget Optimizer.
- Designed scalable PostgreSQL (Prisma) backend for multi-currency expense tracking and settlements.
- **Tech Stack:** Next.js, NestJS, PostgreSQL, Prisma, Tailwind CSS, Groq AI, Mapbox

SyncVerse – Real-Time Collaborative Media Platform Live | GitHub

- Engineered real-time sync system using Socket.IO with sub-second latency for multi-user sessions.
- Built WebRTC P2P network for spatial voice, screen sharing, and media streaming with low server load.
- Developed PostgreSQL (Prisma) backend with Next.js frontend using Zustand and GSAP animations.
- **Tech Stack:** Next.js, Node.js, Socket.IO, WebRTC, PostgreSQL, Prisma, Docker, AWS

PitchPulse – Professional Cricket Management & Live Scoring Platform Live | GitHub

- Designed CQRS-based system for accurate match state tracking and retroactive corrections.
- Built real-time engine using WebSockets and SSE for live scoring and match visualizations.
- Implemented DLS and NRR algorithms for automated match analytics and tournament insights.
- **Tech Stack:** Next.js, Fastify, PostgreSQL, Prisma, Socket.IO, Docker, Kubernetes, AWS, Jenkins

TECHNICAL SKILLS

Languages: Java, JavaScript, Python, C, HTML, CSS
Frontend: React.js, Next.js, Three.js, Tailwind CSS
Backend: Node.js, Express.js, NestJS
Databases: MongoDB, MySQL, PostgreSQL
DevOps & Cloud: Docker, AWS, CI/CD (Jenkins), Kubernetes (Basics)
Tools & Technologies: Git, GitHub, REST APIs, JWT

LEADERSHIP & ACTIVITIES

- **President, Club Innova** – Led 70 members; organized large-scale events with 1000+ attendees.
- **NCC Cadet (A, B & C Certificate Holder)** – Developed leadership, discipline & teamwork through camps.

ACHIEVEMENTS

- Cleared NDA written examination 3 times (AIR 501 in NDA 147)
- Led NCC Naval Unit in multiple training camps